Obstructions to five prescribed Suzuki constructions

Human Compression project

Abstract

The higher-rank constructions in the fixed Suzuki pregeometry article prescribe a full intersection pattern, not just abstract parabolic types. We separate an elementary defect in three printed intersection formulas from the nonexistence of the intended patterns. Two previously established normalizer arguments exclude those patterns and the related first and sixth constructions. The arguments are reproduced here so that the source disposition does not depend on a label in a working ledger.

Write $q = 2^n$, with n odd, and work in the classical Suzuki matrix model. For odd $d \mid n$, let S_d be the standard subgroup obtained by restricting parameters to $\mathbb{F}_{2^d}$. For $d > 1$ this is the simple group $\mathrm{Sz}(2^d)$; $S_1 = \mathrm{Sz}(2)$ has order 20 and is isomorphic to $C_5 : C_4$. Let H_q be the split torus and w its standard involutory normalizer. Put $\Delta_d = H_{2^d}\langle w\rangle$, of order $2(2^d - 1)$, where H_{2^d} denotes the unique subgroup of H_q of order $2^d - 1$.

We retain the classical conjugacy of subfield groups, conjugacy of their maximal split dihedrals, the split-torus centralizer and normalizer, and the involution and Sylow-two structure. These are explicit earlier inputs, not consequences of a finite sample. In the standard coordinates the unique Bruhat parameters give

$$S_a \cap S_b = S_{\gcd(a,b)}. \tag{1}$$

Indeed a matrix belongs to both groups precisely when its uniquely recovered parameters belong to both fields. This also proves the statement when the common field is $\mathbb{F}_2$.

1 Two rigidity arguments

Lemma 1. *For $d > 1$, the normalizer of Δ_d in $\mathrm{Sz}(q)$ is Δ_d. Exactly one conjugate of S_d contains Δ_d.*

Proof. The odd cyclic subgroup of Δ_d is characteristic, and its centralizer is H_q. Every normalizer of Δ_d therefore belongs to $H_q\langle w\rangle$. For $h \in H_q$, conjugation preserves Δ_d if and only if $h^2 \in H_{2^d}$. The quotient H_q/H_{2^d} has odd order, so this is equivalent to $h \in H_{2^d}$. This proves the normalizer claim. If a conjugate of S_d contains Δ_d, pull the latter back into S_d. It is a maximal split dihedral there and can be moved to the standard one by internal conjugacy. After this adjustment the defining conjugator normalizes Δ_d, hence belongs to $\Delta_d \le S_d$. The conjugate is S_d. □

Theorem 2. *Suppose $a, b > 1$ are odd divisors of n. Let S, T be conjugates of S_a, S_b. Suppose one split dihedral subgroup contains maximal split dihedrals of S and T with a common involution. Then S, T are simultaneously conjugate to S_a, S_b. In particular, $S \cap T \cong \mathrm{Sz}(2^{\gcd(a,b)})$.*

Proof. Conjugate the ambient split-torus normalizer to $H_q\langle w\rangle$. Conjugating further by H_q moves the common reflection to w, since squaring is bijective in this odd cyclic torus. Its rotation subgroups of orders $2^a - 1$ and $2^b - 1$ are unique. The two prescribed dihedrals are therefore Δ_a, Δ_b. Lemma 1 forces the two subfield groups to be standard. Now apply (1). □

For the second argument use the standard Sylow-two subgroup U_q and its center $Z_q = \{u(0,t) : t \in \mathbb{F}_q\}$. Its nonidentity central elements are precisely its involutions. Fix $z = u(0,1)$. The retained involution-centralizer theorem says $C_G(z) = U_q$.

Lemma 3. *If a conjugate K of S_d contains z, then the identity together with the involutions in $C_K(z)$ is exactly Z_{2^d}. Consequently, conjugates of S_a, S_b with $\gcd(a,b) = 1$ cannot have a Klein four subgroup in their intersection.*

Proof. The involutions of S_d form one conjugacy class. Adjusting a defining conjugator internally therefore writes $K = S_d^v$ with $v \in C_G(z) = U_q$. Thus $C_K(z) = U_{2^d}^v$. Its identity and involutions form $Z_{2^d}^v = Z_{2^d}$, because U_q centralizes Z_q. If two subfield groups share an involution, first conjugate it to z. Their common commuting involutions lie in $Z_{2^a} \cap Z_{2^b} = Z_{2^{\gcd(a,b)}}$. For coprime degrees this group has order two. A Klein four group would supply three nonidentity commuting involutions, a contradiction. □

2 The rank-three patterns and their extensions

The fixed article L110 reproduces as R4, R5 and R6 three patterns in $\mathrm{Sz}(2^{ab})$, where a, b are distinct odd primes. They require two subfield groups $S \cong \mathrm{Sz}(2^a)$ and $T \cong \mathrm{Sz}(2^b)$ and a split dihedral D with

$$|D| = 2(2^a - 1)(2^b - 1), \quad S \cap D \cong D_{2(2^a-1)}, \quad T \cap D \cong D_{2(2^b-1)}, \quad S \cap T \cap D \cong C_2.$$

They prescribe $S \cap T$ to be, respectively, D_{10}, C_2^2, or C_4. Theorem 2 instead gives $S \cap T \cong S_1$, of order 20. None of these three full patterns exists. The primality conditions on $2^a - 1$ and $2^b - 1$ are not needed for this obstruction.

The pattern R8 has the same coprime subfield degrees and prescribes $S \cap T \cong C_4 \times C_2$. This group contains a Klein four group, so Lemma 3 excludes it. No primality condition on the chosen nonsplit-torus factors is needed for that conclusion.

The implication for a prescribed extension is elementary but must be stated with its hypotheses. If a larger family of actual subgroups has a residue with one of these full intersection patterns, intersecting the family with the residue stabilizer would realize the forbidden pattern. Hence such a larger family cannot exist. This reasoning concerns the full prescribed lattice; a bare list of abstract subgroup types does not specify the required intersections.

L110 Section 3.3.1, printed pp.12–13, derives Constructions 2, 3, 4 and 6 from residues R4, R5, R6 and R8, respectively. The introduction to the constructions on p.15 explicitly requires the full sublattice obtained from the corresponding rank-four pattern. In higher rank, fixing all but the appropriate two subfield types in these prescribed patterns leaves the corresponding rank-three pattern in a subfield stabilizer. The obstruction therefore applies to every intended full lattice in these four constructions, not just to a convenient coordinate placement.

Construction 1 requires a separate check. The rank-four lattice at the end of Section 3.1, printed p.11, has two subfield groups of distinct prime degrees and the split dihedral of order $2(2^a - 1)(2^b - 1)$. Their intersections with that dihedral are their maximal split dihedrals, with a common involution. The two subfield groups are asserted to intersect in D_{10} (the case $n = 5$ in that paragraph). This is exactly the contradiction in Theorem 2; the additional affine parabolic cannot repair it. Construction 1 extends this prescribed rank-four lattice. For more than two subfield types, fixing the other subfield types leaves the same rank-four residue. Thus none of its intended full lattices exists either. Figure 1 is explicitly conditional on existence and flag transitivity; the impossible lattice supplies no realized diagram.

3 The literal formulas are a separate issue

Let the odd field degree be the squarefree product $p_1 \cdots p_m$ and put $n_i = 2^{p_i} - 1$, as in Constructions 2–4. Their printed intersection runs from $j = 0$ through m, so includes $G_m \cong D_{2\prod_i n_i}$. The printed values are D_{10}, $D_4 = C_2^2$, and C_4, respectively. All three are impossible even before any subfield argument. Since each p_i is odd, $5 \nmid 2^{p_i} - 1$; hence the ambient dihedral order is not divisible by 5. Its two-part is exactly 2, so it has no subgroup of order 4. Lagrange's theorem excludes the three printed intersections.

The following source remark instead says that these constructions differ only in the intersection of the Suzuki subgroups. Deleting the last index from the displayed intersection repairs that particular order defect, but does not remove the independent rigidity obstruction to the intended full lattices. We therefore record both conclusions rather than treating an index correction as a proof of existence.

Attribution and scope. The source is D. Leemans, *The residually weakly primitive pregeometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. The subgroup and normalizer premises are classical. The rigidity proofs reproduce the project's earlier Suzuki realization and higher-subfield notes; the contribution here is their explicit application to the first four and sixth higher-rank constructions and the distinction between literal and intended formulas. This is not a classification of all higher-rank Suzuki geometries, all relative positions, or the remaining constructions. It gives no new principal-refutation, open-conjecture, Lean or compression credit.